# MSG No. 223.106 $Pm'\bar{3}'n$ [ Type III, cubic ]

Table 1: Wyckoff site: 2a, site symmetry: m'-3'.

| No. | position | mapping |
|---|---|---|
| 1 | [0, 0, 0] | [1,2,3,4,5,6,7,8,9,10,11,12,37,38,39,40,41,42,43,44,45,46,47,48] |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | [13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36] |

Table 2: Wyckoff site: 6b, site symmetry: m'm'm'..

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | [1,2,3,4,37,38,39,40] |
| 2 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | [5,8,10,12,41,44,46,48] |
| 3 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | [6,7,9,11,42,43,45,47] |
| 4 | $\left[\frac{1}{2}, 0, 0\right]$ | [13,14,21,22,25,26,33,34] |
| 5 | $\left[0, 0, \frac{1}{2}\right]$ | [15,16,23,24,27,28,35,36] |
| 6 | $\left[0, \frac{1}{2}, 0\right]$ | [17,18,19,20,29,30,31,32] |

Table 3: Wyckoff site: 6c, site symmetry: -4m'.2'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, 0, \frac{1}{2}\right]$ | [1,2,13,14,33,34,39,40] |
| 2 | $\left[\frac{3}{4}, 0, \frac{1}{2}\right]$ | [3,4,21,22,25,26,37,38] |
| 3 | $\left[\frac{1}{2}, \frac{1}{4}, 0\right]$ | [5,12,17,19,30,32,44,46] |
| 4 | $\left[0, \frac{1}{2}, \frac{1}{4}\right]$ | [6,9,16,23,27,36,43,47] |
| 5 | $\left[0, \frac{1}{2}, \frac{3}{4}\right]$ | [7,11,15,24,28,35,42,45] |
| 6 | $\left[\frac{1}{2}, \frac{3}{4}, 0\right]$ | [8,10,18,20,29,31,41,48] |

Table 4: Wyckoff site: 6d, site symmetry: -4m'.2'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{2}, 0\right]$ | [1,2,13,14,33,34,39,40] |
| 2 | $\left[\frac{3}{4}, \frac{1}{2}, 0\right]$ | [3,4,21,22,25,26,37,38] |
| 3 | $\left[0, \frac{1}{4}, \frac{1}{2}\right]$ | [5,12,17,19,30,32,44,46] |
| 4 | $\left[\frac{1}{2}, 0, \frac{1}{4}\right]$ | [6,9,16,23,27,36,43,47] |
| 5 | $\left[\frac{1}{2}, 0, \frac{3}{4}\right]$ | [7,11,15,24,28,35,42,45] |
| 6 | $\left[0, \frac{3}{4}, \frac{1}{2}\right]$ | [8,10,18,20,29,31,41,48] |

Table 5: Wyckoff site: 8e, site symmetry: .32'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | [1,5,6,32,34,36] |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | [2,10,11,28,29,33] |
| 3 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | [3,7,12,25,30,35] |
| 4 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | [4,8,9,26,27,31] |

*continued ...*

Table 5

| No. | position | mapping |
|---|---|---|
| 5 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | [13,18,23,39,43,48] |
| 6 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | [14,15,19,40,44,45] |
| 7 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | [16,17,21,38,46,47] |
| 8 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | [20,22,24,37,41,42] |

Table 6: Wyckoff site: 12f, site symmetry: 2m'm'..

| No. | position | mapping |
|---|---|---|
| 1 | $[x, 0, 0]$ | [1,2,39,40] |
| 2 | $[-x, 0, 0]$ | [3,4,37,38] |
| 3 | $[0, x, 0]$ | [5,12,44,46] |
| 4 | $[0, 0, x]$ | [6,9,43,47] |
| 5 | $[0, 0, -x]$ | [7,11,42,45] |
| 6 | $[0, -x, 0]$ | [8,10,41,48] |
| 7 | $\left[\frac{1}{2} - x, \frac{1}{2}, \frac{1}{2}\right]$ | [13,14,33,34] |
| 8 | $\left[\frac{1}{2}, \frac{1}{2}, x + \frac{1}{2}\right]$ | [15,24,28,35] |
| 9 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - x\right]$ | [16,23,27,36] |
| 10 | $\left[\frac{1}{2}, \frac{1}{2} - x, \frac{1}{2}\right]$ | [17,19,30,32] |
| 11 | $\left[\frac{1}{2}, x + \frac{1}{2}, \frac{1}{2}\right]$ | [18,20,29,31] |
| 12 | $\left[x + \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | [21,22,25,26] |

Table 7: Wyckoff site: 12g, site symmetry: 2m'm'..

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, 0, \frac{1}{2}\right]$ | [1,2,39,40] |
| 2 | $\left[-x, 0, \frac{1}{2}\right]$ | [3,4,37,38] |
| 3 | $\left[\frac{1}{2}, x, 0\right]$ | [5,12,44,46] |
| 4 | $\left[0, \frac{1}{2}, x\right]$ | [6,9,43,47] |
| 5 | $\left[0, \frac{1}{2}, -x\right]$ | [7,11,42,45] |
| 6 | $\left[\frac{1}{2}, -x, 0\right]$ | [8,10,41,48] |
| 7 | $\left[\frac{1}{2} - x, 0, \frac{1}{2}\right]$ | [13,14,33,34] |
| 8 | $\left[0, \frac{1}{2}, x + \frac{1}{2}\right]$ | [15,24,28,35] |
| 9 | $\left[0, \frac{1}{2}, \frac{1}{2} - x\right]$ | [16,23,27,36] |
| 10 | $\left[\frac{1}{2}, \frac{1}{2} - x, 0\right]$ | [17,19,30,32] |
| 11 | $\left[\frac{1}{2}, x + \frac{1}{2}, 0\right]$ | [18,20,29,31] |
| 12 | $\left[x + \frac{1}{2}, 0, \frac{1}{2}\right]$ | [21,22,25,26] |

Table 8: Wyckoff site: 12h, site symmetry: 2m'm'..

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, \frac{1}{2}, 0\right]$ | [1,2,39,40] |
| 2 | $\left[-x, \frac{1}{2}, 0\right]$ | [3,4,37,38] |
| 3 | $\left[0, x, \frac{1}{2}\right]$ | [5,12,44,46] |

*continued ...*

Table 8

| No. | position | mapping |
|---|---|---|
| 4 | $\left[\frac{1}{2}, 0, x\right]$ | [6,9,43,47] |
| 5 | $\left[\frac{1}{2}, 0, -x\right]$ | [7,11,42,45] |
| 6 | $\left[0, -x, \frac{1}{2}\right]$ | [8,10,41,48] |
| 7 | $\left[\frac{1}{2} - x, \frac{1}{2}, 0\right]$ | [13,14,33,34] |
| 8 | $\left[\frac{1}{2}, 0, x + \frac{1}{2}\right]$ | [15,24,28,35] |
| 9 | $\left[\frac{1}{2}, 0, \frac{1}{2} - x\right]$ | [16,23,27,36] |
| 10 | $\left[0, \frac{1}{2} - x, \frac{1}{2}\right]$ | [17,19,30,32] |
| 11 | $\left[0, x + \frac{1}{2}, \frac{1}{2}\right]$ | [18,20,29,31] |
| 12 | $\left[x + \frac{1}{2}, \frac{1}{2}, 0\right]$ | [21,22,25,26] |

Table 9: Wyckoff site: 16i, site symmetry: .3.

| No. | position | mapping |
|---|---|---|
| 1 | $[x, x, x]$ | [1,5,6] |
| 2 | $[x, -x, -x]$ | [2,10,11] |
| 3 | $[-x, x, -x]$ | [3,7,12] |
| 4 | $[-x, -x, x]$ | [4,8,9] |
| 5 | $\left[\frac{1}{2} - x, x + \frac{1}{2}, \frac{1}{2} - x\right]$ | [13,18,23] |
| 6 | $\left[\frac{1}{2} - x, \frac{1}{2} - x, x + \frac{1}{2}\right]$ | [14,15,19] |
| 7 | $\left[x + \frac{1}{2}, \frac{1}{2} - x, \frac{1}{2} - x\right]$ | [16,17,21] |
| 8 | $\left[x + \frac{1}{2}, x + \frac{1}{2}, x + \frac{1}{2}\right]$ | [20,22,24] |
| 9 | $\left[x + \frac{1}{2}, \frac{1}{2} - x, x + \frac{1}{2}\right]$ | [25,30,35] |
| 10 | $\left[x + \frac{1}{2}, x + \frac{1}{2}, \frac{1}{2} - x\right]$ | [26,27,31] |
| 11 | $\left[\frac{1}{2} - x, x + \frac{1}{2}, x + \frac{1}{2}\right]$ | [28,29,33] |
| 12 | $\left[\frac{1}{2} - x, \frac{1}{2} - x, \frac{1}{2} - x\right]$ | [32,34,36] |
| 13 | $[-x, -x, -x]$ | [37,41,42] |
| 14 | $[-x, x, x]$ | [38,46,47] |
| 15 | $[x, -x, x]$ | [39,43,48] |
| 16 | $[x, x, -x]$ | [40,44,45] |

Table 10: Wyckoff site: 24j, site symmetry: ..2'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, y, y + \frac{1}{2}\right]$ | [1,33] |
| 2 | $\left[\frac{1}{4}, -y, \frac{1}{2} - y\right]$ | [2,34] |
| 3 | $\left[\frac{3}{4}, y, \frac{1}{2} - y\right]$ | [3,26] |
| 4 | $\left[\frac{3}{4}, -y, y + \frac{1}{2}\right]$ | [4,25] |
| 5 | $\left[y + \frac{1}{2}, \frac{1}{4}, y\right]$ | [5,30] |
| 6 | $\left[y, y + \frac{1}{2}, \frac{1}{4}\right]$ | [6,27] |
| 7 | $\left[-y, y + \frac{1}{2}, \frac{3}{4}\right]$ | [7,28] |
| 8 | $\left[\frac{1}{2} - y, \frac{3}{4}, y\right]$ | [8,29] |
| 9 | $\left[-y, \frac{1}{2} - y, \frac{1}{4}\right]$ | [9,36] |
| 10 | $\left[y + \frac{1}{2}, \frac{3}{4}, -y\right]$ | [10,31] |
| 11 | $\left[y, \frac{1}{2} - y, \frac{3}{4}\right]$ | [11,35] |

*continued ...*

Table 10

| No. | position | mapping |
|---|---|---|
| 12 | $\left[\frac{1}{2}-y,\ \frac{1}{4},\ -y\right]$ | [12,32] |
| 13 | $\left[\frac{1}{4},\ y,\ \frac{1}{2}-y\right]$ | [13,40] |
| 14 | $\left[\frac{1}{4},\ -y,\ y+\frac{1}{2}\right]$ | [14,39] |
| 15 | $\left[-y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$ | [15,42] |
| 16 | $\left[y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$ | [16,43] |
| 17 | $\left[y+\frac{1}{2},\ \frac{1}{4},\ -y\right]$ | [17,44] |
| 18 | $\left[\frac{1}{2}-y,\ \frac{3}{4},\ -y\right]$ | [18,41] |
| 19 | $\left[\frac{1}{2}-y,\ \frac{1}{4},\ y\right]$ | [19,46] |
| 20 | $\left[y+\frac{1}{2},\ \frac{3}{4},\ y\right]$ | [20,48] |
| 21 | $\left[\frac{3}{4},\ -y,\ \frac{1}{2}-y\right]$ | [21,37] |
| 22 | $\left[\frac{3}{4},\ y,\ y+\frac{1}{2}\right]$ | [22,38] |
| 23 | $\left[-y,\ y+\frac{1}{2},\ \frac{1}{4}\right]$ | [23,47] |
| 24 | $\left[y,\ y+\frac{1}{2},\ \frac{3}{4}\right]$ | [24,45] |

Table 11: Wyckoff site: 24k, site symmetry: m’..

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ y,\ z]$ | [1,38] |
| 2 | $[0,\ -y,\ -z]$ | [2,37] |
| 3 | $[0,\ y,\ -z]$ | [3,40] |
| 4 | $[0,\ -y,\ z]$ | [4,39] |
| 5 | $[z,\ 0,\ y]$ | [5,48] |
| 6 | $[y,\ z,\ 0]$ | [6,45] |
| 7 | $[-y,\ z,\ 0]$ | [7,47] |
| 8 | $[-z,\ 0,\ y]$ | [8,46] |
| 9 | $[-y,\ -z,\ 0]$ | [9,42] |
| 10 | $[z,\ 0,\ -y]$ | [10,44] |
| 11 | $[y,\ -z,\ 0]$ | [11,43] |
| 12 | $[-z,\ 0,\ -y]$ | [12,41] |
| 13 | $\left[\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$ | [13,26] |
| 14 | $\left[\frac{1}{2},\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$ | [14,25] |
| 15 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | [15,36] |
| 16 | $\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | [16,35] |
| 17 | $\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | [17,31] |
| 18 | $\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | [18,32] |
| 19 | $\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | [19,29] |
| 20 | $\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | [20,30] |
| 21 | $\left[\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$ | [21,34] |
| 22 | $\left[\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$ | [22,33] |
| 23 | $\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ \frac{1}{2}\right]$ | [23,28] |
| 24 | $\left[z+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$ | [24,27] |

Table 12: Wyckoff site: `48l`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |
| 2 | $[x,\ -y,\ -z]$ | `[2]` |
| 3 | $[-x,\ y,\ -z]$ | `[3]` |
| 4 | $[-x,\ -y,\ z]$ | `[4]` |
| 5 | $[z,\ x,\ y]$ | `[5]` |
| 6 | $[y,\ z,\ x]$ | `[6]` |
| 7 | $[-y,\ z,\ -x]$ | `[7]` |
| 8 | $[-z,\ -x,\ y]$ | `[8]` |
| 9 | $[-y,\ -z,\ x]$ | `[9]` |
| 10 | $[z,\ -x,\ -y]$ | `[10]` |
| 11 | $[y,\ -z,\ -x]$ | `[11]` |
| 12 | $[-z,\ x,\ -y]$ | `[12]` |
| 13 | $\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$ | `[13]` |
| 14 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$ | `[14]` |
| 15 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ x+\frac{1}{2}\right]$ | `[15]` |
| 16 | $\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$ | `[16]` |
| 17 | $\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | `[17]` |
| 18 | $\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | `[18]` |
| 19 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | `[19]` |
| 20 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[20]` |
| 21 | $\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$ | `[21]` |
| 22 | $\left[x+\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$ | `[22]` |
| 23 | $\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ \frac{1}{2}-x\right]$ | `[23]` |
| 24 | $\left[z+\frac{1}{2},\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$ | `[24]` |
| 25 | $\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$ | `[25]` |
| 26 | $\left[x+\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$ | `[26]` |
| 27 | $\left[z+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-x\right]$ | `[27]` |
| 28 | $\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$ | `[28]` |
| 29 | $\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[29]` |
| 30 | $\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | `[30]` |
| 31 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | `[31]` |
| 32 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | `[32]` |
| 33 | $\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$ | `[33]` |
| 34 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$ | `[34]` |
| 35 | $\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ x+\frac{1}{2}\right]$ | `[35]` |
| 36 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$ | `[36]` |
| 37 | $[-x,\ -y,\ -z]$ | `[37]` |
| 38 | $[-x,\ y,\ z]$ | `[38]` |
| 39 | $[x,\ -y,\ z]$ | `[39]` |
| 40 | $[x,\ y,\ -z]$ | `[40]` |
| 41 | $[-z,\ -x,\ -y]$ | `[41]` |
| 42 | $[-y,\ -z,\ -x]$ | `[42]` |
| 43 | $[y,\ -z,\ x]$ | `[43]` |
| 44 | $[z,\ x,\ -y]$ | `[44]` |
| 45 | $[y,\ z,\ -x]$ | `[45]` |
| 46 | $[-z,\ x,\ y]$ | `[46]` |

*continued ...*

Table 12

| No. | position | mapping |
|---|---|---|
| 47 | $[-y,\ z,\ x]$ | `[47]` |
| 48 | $[z,\ -x,\ y]$ | `[48]` |